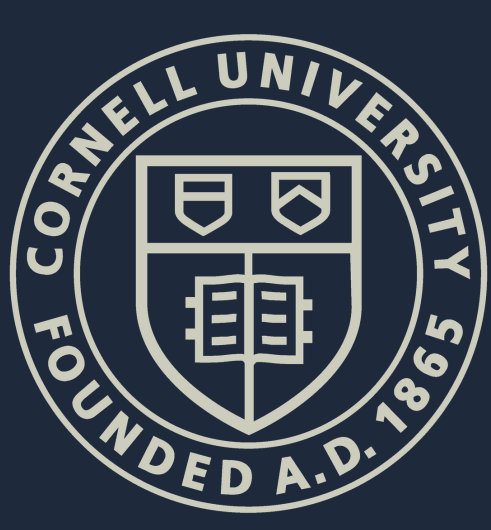




RAGs to Riches:

A Reimplementation of Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks
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Introduction

- **Closed-book models** such as T5 rely *only* on memorized knowledge, which can make them *unreliable* for knowledge-intensive QA and prone to *hallucination*.
- **Retrieval-Augmented Generation (RAG)** improves this by *retrieving* relevant evidence *before* generation, combining parametric and non-parametric memory.
- **Goal:** evaluate whether external retrieval improves factual question answering compared with closed-book baselines in open-domain QA.

Background

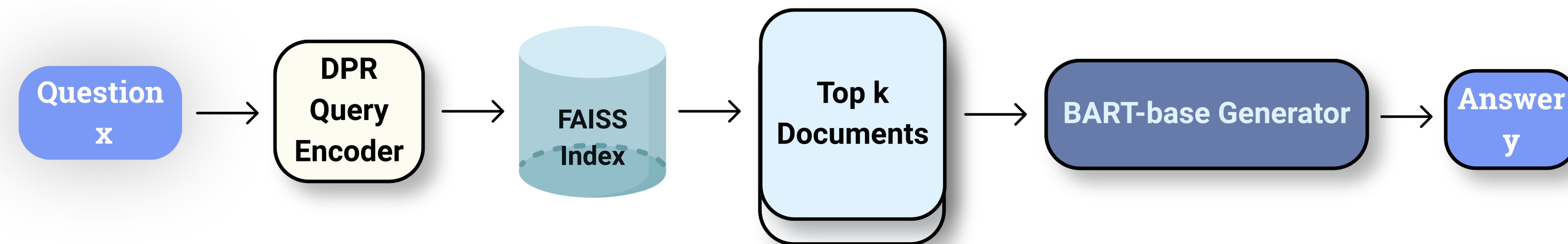
- Lewis et al. introduced RAG-Token and RAG-Sequence as two retrieval-augmented variants for open-domain QA.
- **Reproduced Result:** a smaller scale version of Table 1. RAG variants vs. T5 and DPR baselines on Natural Questions.

Methodology

- **Models & Tools:** DPR retriever + BART-base generator + FAISS passage index.
- **Dataset:** Natural Questions (NQ), 1,000-sample subset due to compute constraints.
- **Modifications:** SQuAD instead of Wiki-DPR due to availability; BART-base instead of BART-large due to memory limits.
- **Metrics:** Token-level F1 instead of Exact Match.
- **Additional Experiment:** Tested retrieval depth $k \in \{1, 3, 5, 10\}$ on both RAG variants.

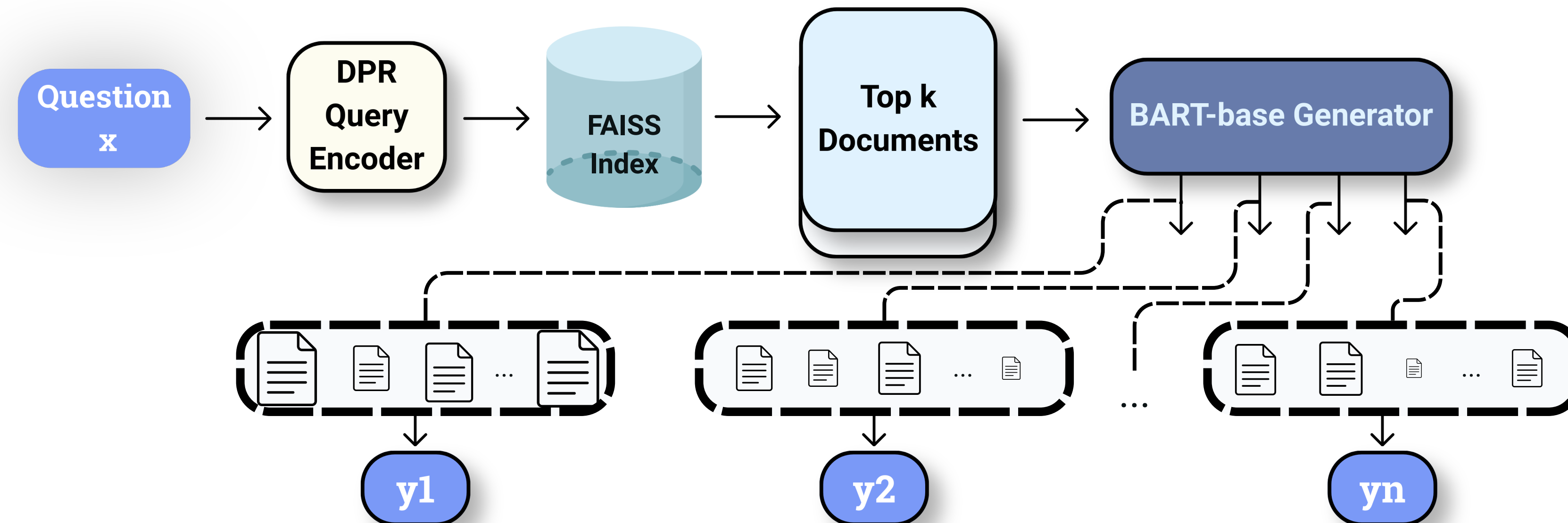
RAG-Sequence

Best at forming long coherent answers



RAG-Token

Best for brief fact generation where you need answers from within multiple documents



Results

Natural Questions (NQ): Paper vs. Our Reproduction

Book	Model	Paper NQ (EM)	Our NQ (F1)
Closed Book	T5 (paper: T5-11B, ours: T5-base)	34.5	8.1
Open Book	DPR	41.5	13.73
Open Book	RAG-Token	44.1	33.8
Open Book	RAG-Sequence	44.5	31.44

Figure 1. Smaller-scale NQ reproduction. Our scores use token-level F1, while the original paper reports Exact Match.

Effect of Retrieval Depth (k)

k	RAG-Token F1	RAG-Sequence F1
1	32.1	31.48
3	33.92	31.48
5	33.8	31.48
10	33.89	31.48

Figure 2. Increasing retrieval depth improves RAG-Token, while RAG-Sequence remains nearly unchanged in our setup.

Conclusion

- **Retrieval Helps:** RAG performed better than the closed-book T5 baseline, showing that using outside evidence helps models answer factual open-domain questions more accurately.
- **Token vs. Sequence:** RAG-Token handled imperfect retrieval better, while RAG-Sequence depended more on one strong set of retrieved documents and struggled when retrieval quality was low.
- **Effect of k:** Increasing retrieval depth gave RAG-Token more evidence to use, but the small gains suggest that after a point, extra passages add limited useful information.

Future Work

- **Corpus Scaling:** Scale our reproduction to the full Wikipedia/Wiki-DPR corpus and larger generators to better match the original paper.
- **Retrieval Upgrades:** Improve retrieval with stronger or newer retrievers, since evidence quality is the main bottleneck.
- **Contrastive Grounding:** Use negative retrieval samples during fine-tuning to penalize the generator for ungrounded outputs.

References

- [1] Lewis, P., et al. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." *Advances in Neural Information Processing Systems*, vol. 33, 2020